

Fitness Unlimited Schema Integration and Design

Joshua Schwenk

Colorado State University Global

MIS407: Database Concepts

Dr. Shawn Harris

8/24/2025

Fitness Unlimited Schema Integration and Design

Fitness Unlimited is a rapidly growing provider of exercise centers that offers memberships, merchandise, and specialized services. Their current data environment is fragmented across operational systems and spreadsheets. Merchandise transactions, service purchases, and special events are tracked separately which limits the company's ability to see a complete picture of performance. To support scalability and strategic decision-making, Fitness Unlimited needs a unified data warehouse that enables analysis across franchise, product/service type, customer, and time.

This proposal presents a complete solution to Fitness Unlimited's schema integration and design. Using their operational retail database and the special events spreadsheet, I identify dimensional model components, specify measures with aggregation properties, determine the grain and estimated storage, and design a constellation schema for inventory analysis.

1. Facts, Dimensions, Data-Source Mapping, and Hierarchies

Table 1

Fact Tables, Data Sources, and Attributes

Fact Table	Data Source	Attributes
FactMerchSales	Sales, Contains, Merchandise	SaleDate, FranchId, MmbrId, MerchId, Qty, MerchPrice
FactServicePurchases	ServicePurchase, ServiceCategory	ServPurchDate, FranchId, MmbrId, ServCatId, ServCatPrice
FactSpecialEventsSales	Special Events Worksheet	EventDate, FranchId, CorporateCustomerId, EventTypeCode, Amount

Note. Fact tables capture transactional data at the most detailed level of analysis, with one row per merchandise sale, service purchase, or special event. Attributes include identifiers, dates, and values that support flexible aggregation and reporting.

Table 2

Dimension Mapping to Data Sources and Hierarchies

Dimension	Data Source	Attributes	Hierarchy
DimDate	Sale.SaleDate, ServPurchase.Date, SpecialEvents.Date	Datekey, FullDate, Year, Quarter, Month, Week, DayName	Year > Quarter > Month > Week > Date
DimFranchise	Franchise	FranchId, FranchRegion, FranchPostalCode, FranchModelType	FranchRegion > FranchPostalCode > FranchID
DimMember	Member, MemberType	MmbrId, MmbrName, MmbrZip,	MemberTypeName > MmbrDate > MmbrId

		MemTypeName, MemTypePrice, MmbrDate, Email	
DimMerchandise	Merchandise	MerchId, MerchName, MerchType, MerchPrice	MerchType > MerchName
DimServiceCategory	ServiceCatagory	ServCatId, ServCatName, ServCatPrice	Flat (nonhierarchical)
DimCorporateCustomer	Special Events Worksheet	CorpCustId, OrgName, Location	OrgName > Location > CorpCustId
DimEventType	Special Events Worksheet	EventTypeCode, EventName	EventTypeCode > EventName

Note. Dimension tables provide descriptive context for facts and enable roll-up analysis. Hierarchies are listed from broad to narrow levels. Flat indicates nonhierarchical dimensions where only a single level exists (Mannino, 2022).

2. Measures, Sources, and Aggregation Properties

Measures represent the quantitative data that Fitness Unlimited will analyze across dimensions. The relative measures, their sources, and aggregation behaviors are summarized in Table 3.

Table 3

Measures, Data Sources, and Aggregation Properties

Measure	Data Source	Aggregation Properties
Quantity (Qty)	Contains.Qty, ServPurchase	Fully additive across all dimensions.
MerchPrice	Merchandise.MerchPrice	Non-additive; used for descriptive detail or averages.
ServCatPrice	ServiceCategory.ServCatPrice	Non-additive; used for descriptive detail or averages.
Amount	SpecialEvents.Amount	Additive across events, franchises, and dates.
SalesAmount	Derived (Qty x Price/Amount)	Fully additive; primary measure for revenue analysis.
TransactionCount	Derived (row count)	Additive; useful for averages.

Note. Measures represent the quantitative values used for analysis. Quantities are fully additive and allow aggregation across all dimensions. Prices are non-additive and should not be summed. Instead, averages can be calculated by dividing SalesAmount by total quantity. Amount values from special events are stored directly and are additive. SalesAmount and TransactionCount are derived measures created to support revenue and trend analysis.

3. Grain, Cardinality, and Storage Estimates

The grain of the Fitness Unlimited constellation schema is defined at the transactional level for each fact. For merchandise, each row in FactMerchSales represents one line item within a sale. For services, each row in FactServicePurchases represents one unit of service purchased. For special events, each row in FactSpecialEventsSales corresponds to one promotional event sold to corporate customer. Defining the grain at this detail level ensures that the data warehouse can flexibly support aggregation across franchise, customer, and product/service dimensions.

Cardinality estimates from the operational systems indicate that Fitness Unlimited processes approximately 450,000 merchandise line items, 100,000 service purchases, and 60,000 special events per year. This results in a combined fact table size of roughly 610,000 rows annually.

To estimate storage, each row requires both the raw attribute data and overhead. MySQL (2025) documentation shows that rows often occupy more space than the size of their explicit attributes. A conservative estimate of around 50 bytes per row accounts for both attributes storage and system overhead. Using this estimate, annual storage for 610,000 rows equals to roughly 30.5 MB. Even over multiple years, this remains relatively low in comparison to modern enterprise data warehouse capacity.

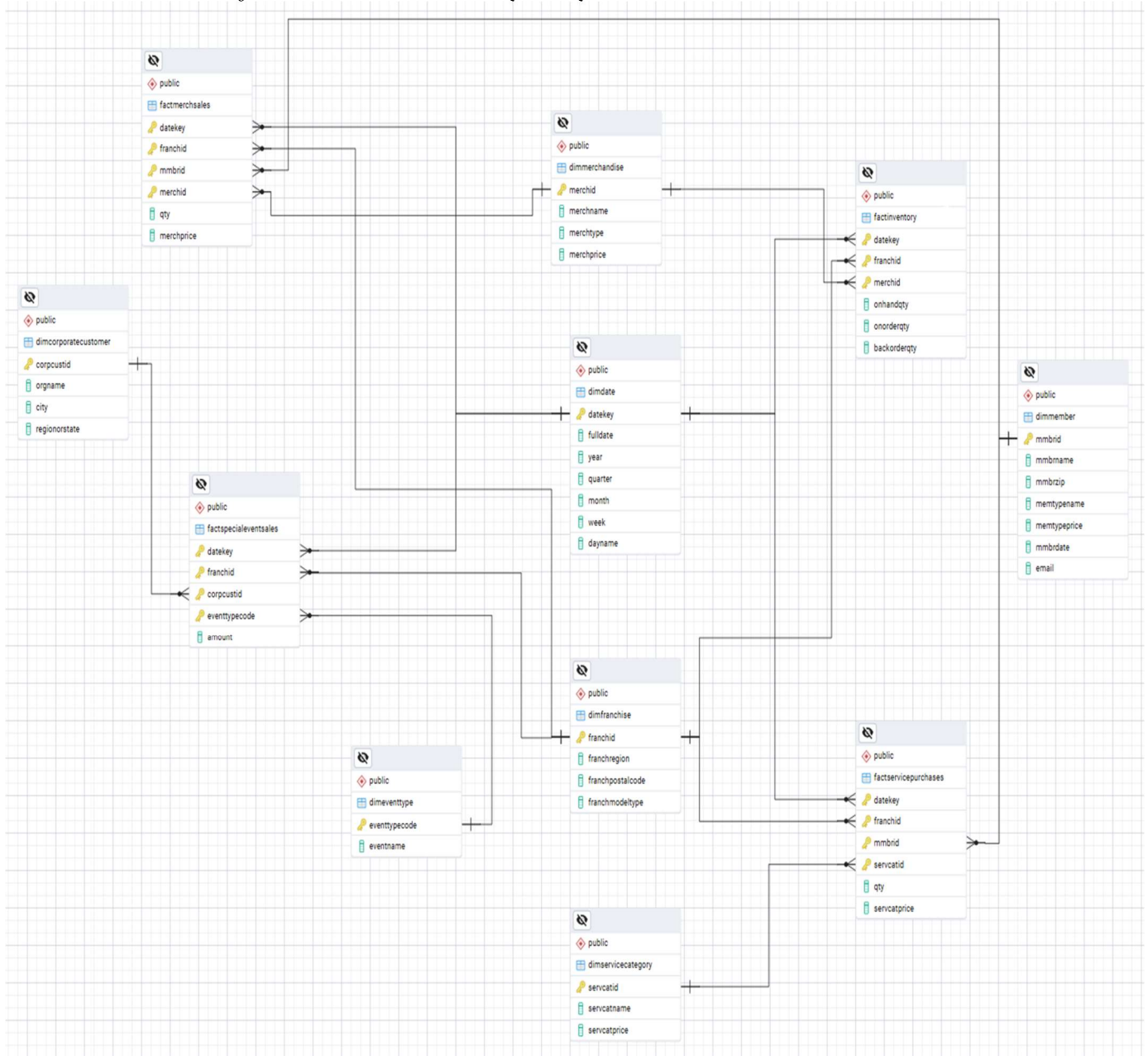
The fact tables are also sparse relative to the full set of possible dimensional combinations. For example, if every franchise, customer, merchandise/service item, and date were represented, the total potential rows would be in the millions. However, the warehouse does not store empty combinations. Instead, rows are recorded only for actual sales, service purchases, or special events, so sparsity does not inflate storage requirements (Crampton, 2022).

4. Inventory Analysis: Constellation Schema

Figure 1 shows the extended constellation schema that incorporates inventory analysis alongside existing revenue facts. The new table, FactInventory, captures periodic snapshots of merchandise stock at the grain of DateKey, FranchId, and MerchId. Its primary key is the composite of Date, Franchise, and Merchandise. Its columns include OnHandQty, OnOrderQty, and BackorderQty. By sharing DimDate, DimFranchise, and DimMerchandise, with sales facts, the warehouse enables cross fact analysis such as comparing weekly sales with week-ending inventory. Inventory measures are semi-additive, meaning quantities can be summed across franchises or merchandise, but not across dates. This design applies flattening of franchise and merchandise attributes to simplify rollups and ensure the constellation supports both sales and inventory perspectives within a single integrated model.

Figure 1

Constellation Schema for Revenue and Inventory Analysis



Note. This diagram shows four fact tables (FactMerchSales, FactServicePurchases, FactSpecialEventsSales, and FactInventory) connected to shared conformed dimensions. FactInventory extends the constellation to support analysis of on-hand, on-order, and backorder quantities.

References

Mannino, M. (2022). *Database design: Design, query, formulation, and administration using Oracle and PostgreSQL*. SAGE Publications.

MySQL. (2025). Storage requirements for table columns. *MySQL 8.3 reference manual*. Oracle. <https://dev.mysql.com/doc/refman/8.3/en/storage-requirements.html>

Crampton, A. (2022, February 17). *Everything you need to know about snapshot fact tables*. Telefonica Tech. <https://telefonicatech.uk/blog/everything-you-need-to-know-about-snapshot-fact-tables/>